



ExoHabit-AI

Exploring the Cosmos, One Exoplanet at a Time



What is ExoHabit-AI?

- An intelligent web application that predicts exoplanet habitability using machine learning
- Analyzes 10 key planetary and stellar parameters to determine if a planet could support life
- Provides instant predictions through an interactive web interface
- Educational platform featuring famous exoplanets and discovery methods
- Built with Flask backend and immersive 3D frontend experience
- Deployed and accessible online at <https://exohabit-ai-3ni1.onrender.com>

BACKGROUND

What Are Exoplanets?

- Planets that orbit stars beyond our Solar System
- Over 5,600 confirmed exoplanets discovered to date
- Most orbit other stars, but some are 'rogue planets' floating freely
- Discovery methods include: Transit, Radial Velocity, Direct Imaging, Microlensing
- The 'Habitable Zone' or 'Goldilocks Zone' is where liquid water could exist
- NASA's Kepler and TESS missions have revolutionized discovery

Discovery Timeline

- 1992: First exoplanets discovered orbiting pulsar PSR B1257+12
- 1995: 51 Pegasi b - First exoplanet around a sun-like star (Hot Jupiter)
- 2009: Kepler Space Telescope launches, discovers thousands of planets
- 2015: TRAPPIST-1 system discovered with seven Earth-sized planets
- 2018: TESS mission continues the search for nearby exoplanets
- 2021: James Webb Space Telescope begins detailed atmosphere analysis

Key Features

ML Prediction

Advanced Random Forest model predicts habitability with high accuracy

3D Visualization

Interactive 3D starfield using Three.js technology

Real-time API

RESTful API for instant predictions and batch processing

Educational Content

Learn about famous exoplanets and discovery methods

Responsive Design

Mobile-friendly interface with beautiful animations

Data Visualization

Interactive charts showing planetary data and statistics

Technology Stack

Backend

- Python 3.8+
- Flask Web Framework
- Scikit-learn ML
- Random Forest Algorithm
- Joblib Serialization
- Pandas/NumPy
- Flask-CORS

Frontend

- HTML5/CSS3
- JavaScript ES6+
- Three.js 3D Graphics
- Chart.js Visualizations
- GSAP Animations
- Responsive Design
- Orbitron/Exo 2 Fonts

Machine Learning Model

Model Details

- Algorithm: Random Forest Classifier
- Trained on NASA Exoplanet Archive data
- 10 input features analyzed
- Binary classification: Habitable vs Not Habitable
- Probability scores for confidence levels
- Joblib serialization for fast loading

Input Features

- pl_orbper: Orbital period (days)
- pl_rade: Planet radius (Earth radii)
- pl_bmasse: Planet mass (Earth masses)
- pl_eqt: Equilibrium temperature (K)
- st_teff: Stellar temperature (K)
- st_rad: Stellar radius (Solar radii)
- st_mass: Stellar mass (Solar masses)
- sy_dist: Distance to system (parsec)
- sy_snum: Number of stars
- sy_pnum: Number of planets

REST API Endpoints

GET

/

API information and status

GET

/health

Health check endpoint

GET

/features

List of required input features

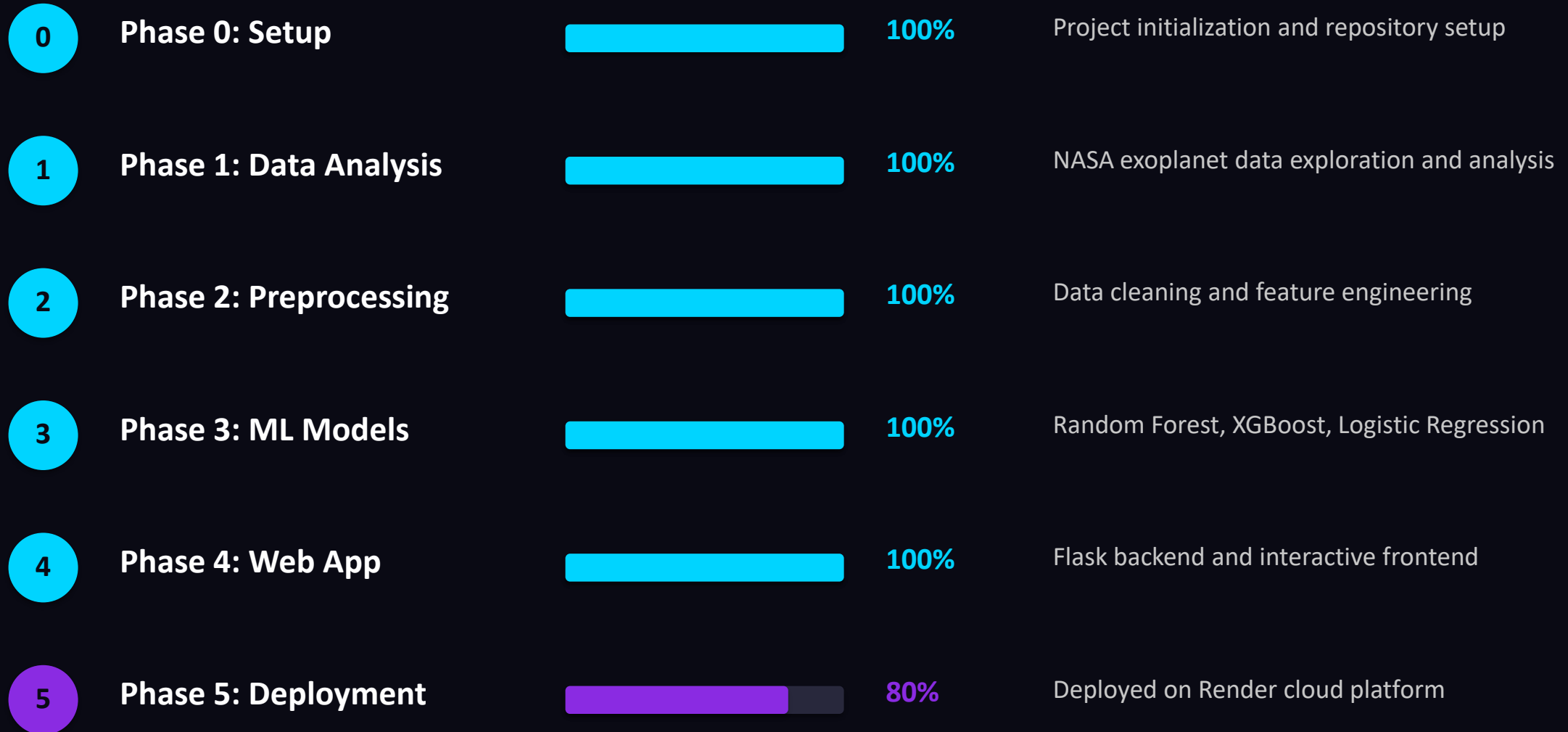
POST

/predict

Single exoplanet habitability prediction

```
Example Response: {"success": true, "prediction": {"is_habitable": 1, "habitability_probability": 0.85, "confidence": "High", "classification": "Habitable"}}
```


Project Roadmap



Exoplanet Statistics

5,600+

Confirmed Exoplanets

205

Potentially Habitable

4.3

Light Years to Proxima b

30+

Discovery Methods

1992

First Exoplanets Found

10

Input Features Analyzed

Use Cases & Applications

Research & Education

- Students learning about exoplanets
- Astronomy enthusiasts exploring data
- Quick habitability assessments
- Understanding planetary parameters
- Interactive learning experience

API Integration

- Third-party app integration
- Batch analysis of multiple planets
- Data science projects
- Research tool automation
- Custom visualization tools

Future Enhancements

- Integration with real-time NASA Exoplanet Archive API
- Advanced atmospheric composition analysis
- Machine learning model improvements with more data
- Mobile app development for on-the-go exploration
- Community features for sharing discoveries
- Multi-language support for global accessibility

Acknowledgments

- NASA Exoplanet Archive for providing comprehensive exoplanet data
- Kepler and TESS missions for discovering thousands of exoplanets
- Open Source community for Flask, Scikit-learn, Three.js, and other libraries
- Springboard Mentor program for project guidance and support
- Contributors and collaborators who helped build this project



Thank You!

Explore the cosmos with ExoHabit-AI

Live Demo: <https://exohabit-ai-3ni1.onrender.com>

GitHub: <https://github.com/springboardmentor1234r/ExoHabit-AI>